

RISC-V MCU development boards

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Introduction

"How do I get my feet wet with RISC-V?" is a very common question, which is often answered "Buy a RISC-V development board, and practice bare metal development."

This document provides guidance on part selection, as well as pointers to useful resources. It is not a comprehensive guide of RISC-V MCU, but rather a short list of easily approachable parts. To be listed in this guide, parts **must** meet several criteria:

- have decent English documentation,
- have open-source or freely downloadable supporting software,
- be easily available from anywhere in the world, e.g. through AliExpress or LCSC,
- have cheap development boards as easily available as the chips,
- not require a Chinese mobile phone number to download software or documentation.

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Manufacturer selection

MCU manufacturers have different product strategies, leading to different product ranges. An easy and efficient method is to select a manufacturer whose strategy matches your needs, and then see which of their parts best suits your project.

<i>Manufacturer</i>	<i>Wide supply voltage range</i>	<i>Wide package choice</i>	<i>64-bit</i>	<i>Lots of I/O pins</i>	<i>WiFi</i>	<i>BLE</i>	<i>802.15.4</i>
BouffaloLab			x		x	x	x
Espressif					x	x	x
GigaDevice				x			
SOPHGO			x				
WCH	x	x		x		x	

Note: manufacturers are sorted in alphabetical order.

The table above only lists chips manufacturers offering an easy experience to people living outside China. However, there are also interesting domestic RISC-V MCU, described at the end of this document. Refer to the chapter "**Before you read on, a few words of warning**" for more about the difficulties you should expect.

BouffaloLab

Chinese name: 博流智能 (or just 博流) – pinyin: bó liú zhì néng – <https://en.bouffalolab.com/>

BouffaloLab only manufactures RISC-V chips, so they're undoubtedly committed to this platform. They have a wide range of IoT-oriented MCU, some with SiFive IP, others with T-Head IP. The documentation is good, but their SDK and code examples are a bit chaotic.

2024-02-05: BouffaloLab is working on a unified SDK (bouffalo_sdk) to replace the old bl_mcu_sdk and bl_iot_sdk. However, their works progresses intolerably slowly. After many months, wireless is still only supported for the BL616/618, and support for many peripherals is still marked as “unsupported” or “supported but not tested” (sic). I wonder how they can possibly sell chips without a proper SDK.

The following table presents the main differences between BouffaloLab's RISC-V SoC (not the modules):

Part	MHz	RAM	CPU	Core IP	WiFi	BT	BLE	802.15.4	USB	Ethernet	Camera	AI NPU
BL602/ BL604	192	276K	RV32IMAFRC	SiFive E24	b/g/n	-	yes	-	-	-	-	-
BL702	144	132K	RV32IMAFRC	SiFive E24	-	-	yes	yes	yes	-	-	-
BL704	144	132K	RV32IMAFRC	SiFive E24	-	-	yes	yes	yes	yes	-	-
BL706	144	132K	RV32IMAFRC	SiFive E24	-	-	yes	yes	yes	yes	yes	-
BL616/ BL618 *	320	480K	RV32IMAFRC	T-Head E907	b/g/n/ax	yes	yes	yes	yes	yes	yes	-
BL808	480	64M	RV32IMAFRC+ RV64GCV+ RV32EMC	T-Head E907 + C906 + E902	b/g/n	yes	yes	yes	yes	yes	yes	yes

*: the BL616 and BL618 are ultra-low-power wireless MCU.

I chose to include the BL808 because it is exceedingly powerful for an MCU, but very limited for a Linux-capable SoC, so it makes sense to include it in both documents.

I didn't include the BL606P because as of 2023-06-27, BouffaloLab provides no technical documentation about it, and neither chips, nor development boards are available for purchase.

Finally, the BL604, is a higher pin count version of the BL602, and the BL618, a higher pin count version of the BL616.

Clones

The BL602 is so successful that it has clones, the LF686 (= BL602) and LF688 (= BL604) by [LeapFive](#), and the TG7100C (= BL602) by [T-Head](#) (developed for [Tmall](#)).

Advice to beginners

Pine64 has released a well-documented development board (PineCone) and a module (PineNut), both based on the BL602 MCU, and [Lee Lup Yuen](#) has produced [excellent training material](#) for this board, so this is what you want to use for your first steps.

Energy efficiency

Since the early days of the BL602, BouffaloLab has always paid attention to energy efficiency. This commitment is confirmed today by the priority they give to the BL616/618, their ultra-low-power MCU. If they meet your needs, these are the MCU you want to focus on.

Compute power & embedded AI

The BL808 is a great choice for applications that need significant compute power while staying energy efficient, and for applications incorporating AI-based features.

Documentation, SDK and tools

Data sheets & reference manuals: https://github.com/bouffalolab/bl_docs

SDK:

https://github.com/bouffalolab/bouffalo_sdk (newer SDK)

https://github.com/bouffalolab/bl_iot_sdk (older SDK)

Note: the older SDK may still be needed until the newer has full peripheral coverage (e.g. to use WiFi on the BL808).

<https://dev.bouffalolab.com/download>

Flashing tool (BIDevCube): https://github.com/bouffalolab/flash_tools

GitHub: <https://github.com/bouffalolab>

Developer forum: <https://bbs.bouffalolab.com/t/english-forum>

OpenBouffalo wiki, covers many practical details: <https://openbouffalo.org/>

Interesting information not found in BouffaloLab documents: <https://github.com/pine64/>

BL602

Third-party development board: Pine64 PineCone

Purchase link: <https://pine64.com/product/pinecone-bl602-evaluation-board/>

Documentation: <https://wiki.pine64.org/wiki/PineCone>

Review: <https://lupyuen.github.io/articles/pinecone>

Tutorials: <https://lupyuen.github.io/articles/book>

Third-party development board: Ai-Thinker Ai-WB2 series

Purchase links (Ai-WB2-12F-Kit: 11 I/O):

<https://www.aliexpress.com/item/1005004911487557.html>

<https://www.aliexpress.com/item/1005005287884896.html>

<https://www.aliexpress.com/item/1005005742607807.html>

<https://www.aliexpress.com/item/1005005256873547.html>

Purchase links (Ai-WB2-13-Kit: 11 I/O):

<https://www.aliexpress.com/item/1005004911837141.html>

<https://www.aliexpress.com/item/1005005287884896.html>

<https://www.aliexpress.com/item/1005005256844150.html>

<https://www.aliexpress.com/item/1005005699841474.html>

Purchase links (Ai-WB2-32S-Kit: 15 I/O):

<https://www.aliexpress.com/item/1005004911645385.html>

<https://www.aliexpress.com/item/1005005287884896.html>

<https://www.aliexpress.com/item/1005005256875832.html>

<https://www.aliexpress.com/item/1005005697279606.html>

Documentation: <https://docs.ai-thinker.com/en/wb2>

GitHub: <https://github.com/Ai-Thinker-Open/Ai-Thinker-WB2>

Third-party development board: DT-BL10 or XT-BL12

Purchase links:

<https://www.aliexpress.com/item/1005005083839351.html>

<https://www.aliexpress.com/item/1005001762587381.html>

<https://www.aliexpress.com/item/1005003695650307.html>

<https://www.aliexpress.com/item/1005004477041228.html>

Documentation: <http://bbs.doit.am/forum.php?mod=viewthread&tid=482>

BL702

Third-party development board: Sipeed M0 Sense

Purchase links:

<https://www.aliexpress.com/item/1005005373072135.html>

<https://www.aliexpress.com/item/1005005686395980.html>

<https://www.aliexpress.com/item/1005005363184503.html>

<https://www.aliexpress.com/item/1005005372923816.html>

<https://www.aliexpress.com/item/1005005012406688.html>

Documentation: <https://dl.sipeed.com/shareURL/Maix-Zero/M0sense>

GitHub: https://github.com/sipeed/M0sense_BL702_example

Third-party development board: XiaoTeng Technology XT-ZB1

Purchase links:

<https://www.aliexpress.com/item/1005004477055377.html>

<https://www.aliexpress.com/item/1005003695882418.html>

<https://www.aliexpress.com/item/1005003747200098.html>

<https://www.aliexpress.com/item/1005004134568356.html>

<https://www.aliexpress.com/item/1005004705201239.html>

Documentation: <http://bbs.doit.am/forum.php?mod=viewthread&tid=488>

Third-party development board: Shenzhen LC Technology 303BL702002 = BL702S

Purchase links:

<https://www.aliexpress.com/item/1005005927910587.html>

<https://www.aliexpress.com/item/1005005880754314.html>

<https://www.aliexpress.com/item/1005005880723515.html>

<https://www.aliexpress.com/item/1005005848279255.html>

<https://www.aliexpress.com/item/1005006020089592.html>

BL616

Third-party development board: Sipeed M0S Dock

Purchase links:

<https://www.aliexpress.com/item/1005005373075939.html>
<https://www.aliexpress.com/item/1005005286453236.html>
<https://www.aliexpress.com/item/1005005261055758.html>
<https://www.aliexpress.com/item/1005005743601410.html>
<https://www.aliexpress.com/item/1005005142466936.html>

Documentation: <https://dl.sipeed.com/shareURL/Maix-Zero/M0S>

GitHub: https://github.com/sipeed/M0S_BL616_example

Third-party development board: Ai-Thinker Ai-M62-12F

Purchase links:

<https://www.aliexpress.com/item/1005005553858124.html>
<https://www.aliexpress.com/item/1005005742683460.html>
<https://www.aliexpress.com/item/1005005407942430.html>
<https://www.aliexpress.com/item/1005005438854506.html>

Documentation: https://docs.ai-thinker.com/en/ai_m62

GitHub: https://github.com/Ai-Thinker-Open/aithinker_Ai-M6X_SDK

BL618

Third-party development board: Sipeed M0P Dock

Purchase links:

<https://www.aliexpress.com/item/1005005505353135.html>
<https://www.aliexpress.com/item/1005005505242737.html>
<https://www.aliexpress.com/item/1005005434411547.html>
<https://www.aliexpress.com/item/1005005461103465.html>

Documentation: <https://dl.sipeed.com/shareURL/Maix-Zero/M0P>

GitHub: https://github.com/sipeed/M0P_BL618_examples

Third-party development board: Ai-Thinker Ai-M61-32S

Purchase links:

<https://www.aliexpress.com/item/1005004486335583.html>
<https://www.aliexpress.com/item/1005005525538426.html>
<https://www.aliexpress.com/item/1005005407539968.html>
<https://www.aliexpress.com/item/1005005407935386.html>

Documentation: https://docs.ai-thinker.com/en/ai_m61

GitHub: https://github.com/Ai-Thinker-Open/aithinker_Ai-M6X_SDK

BL808

Third-party development board: Sipeed M1S Dock

Purchase links:

<https://www.aliexpress.com/item/1005004996572935.html>

<https://www.aliexpress.com/item/1005004996668405.html>

<https://www.aliexpress.com/item/1005004996731092.html>

<https://www.aliexpress.com/item/1005004970779483.html>

Documentation: <https://dl.sipeed.com/shareURL/MAIX/M1s>

GitHub:

https://github.com/sipeed/M1s_BL808_example

https://github.com/sipeed/M1s_BL808_SDK

https://github.com/sipeed/M1s_BL808_Linux_SDK

Release date: Dec. 2022

Third-party development board: Pine64 Ox64

Purchase link: <https://pine64.com/product-category/ox64/>

Documentation:

<https://wiki.pine64.org/wiki/Ox64>

<https://lupyuen.codeberg.page/articles/ox64.html>

<https://lupyuen.codeberg.page/articles/ox2.html>

<https://lupyuen.codeberg.page/articles/mmu.html>

<https://lupyuen.codeberg.page/articles/app.html>

<https://lupyuen.codeberg.page/articles/plic2.html>

<https://lupyuen.codeberg.page/articles/plic3.html>

<https://www.hackster.io/lupyuen/8-risc-v-sbc-on-a-real-time-operating-system-ox64-nuttx-474358>

Release date: Dec. 2022

Espressif

Chinese name: 乐鑫科技 – Pinyin: lè xīn kējì – <https://www.espressif.com/>

Espressif is famous for their Xtensa-based ESP32 and ESP32-S3 modules, but they also announced their new developments would be made on RISC-V, so interesting things are to be expected from them. Like BouffaloLab, they focus on IoT-oriented MCU.

Their current RISC-V product range includes the ESP32-C2/ESP8684, ESP32-C3/ESP8685, ESP32-C6, and ESP32-H2.

The following table presents the main differences between Espressif's RISC-V SoC (not the modules):

Part	Pins	CPU	MHz	RAM	Max. flash	WiFi	BLE	802.15.4	Notes
ESP8684	24	RV32IMC	120	272K	4M	b/g/n	yes	-	Aka. ESP32-C2
ESP32-C3	32	RV32IMC	160	400K	4M	b/g/n	yes	-	
ESP8685	28	RV32IMC	160	400K	4M	b/g/n	yes	-	
ESP32-C6	40	RV32IMAC	160	512K	4M	b/g/n/ax	yes	yes	Has 2 CPU (one low power)
ESP32-H2	32	RV32IMAC	96	320K	4M	-	yes	yes	Low-power applications

GitHub: <https://github.com/espressif>

The SDK for all Espressif products is called **ESP-IDF**. It is available from GitHub, and the latest stable version can be downloaded from here:

[https://www.espressif.com/en/support/download/all?field_type_tid\[\]=785](https://www.espressif.com/en/support/download/all?field_type_tid[]=785)

ESP8684 (ESP32-C2)

Official development board: ESP8684-DevKitM-1

Purchase link: <https://www.aliexpress.com/item/1005004436990376.html>

Documentation: <https://docs.espressif.com/projects/espressif-esp-dev-kits/en/latest/esp8684/esp8684-devkitm-1/>

Official development board: ESP8684-DevKitC-02

Purchase link: <https://www.aliexpress.com/item/1005004693162839.html>

Documentation: <https://docs.espressif.com/projects/espressif-esp-dev-kits/en/latest/esp8684/esp8684-devkitc-02/>

Third-party development board: NODEMCU-ESP32-C2

Purchase links:

<https://www.aliexpress.com/item/1005006043762444.html>

<https://www.aliexpress.com/item/1005005939171867.html>

<https://www.aliexpress.com/item/1005005946826508.html>

<https://www.aliexpress.com/item/1005006009378965.html>

<https://www.aliexpress.com/item/1005005939349465.html>

Compatible with ESP8684-DevKitM-1.

Third-party development board: ESPC2-12

Purchase link: <https://www.aliexpress.com/item/1005004708803007.html>

Documentation: <http://bbs.doit.am/forum.php?mod=viewthread&tid=489>

Third-party development board: ESP32-C2 DEVKIT

Purchase links:

<https://www.aliexpress.com/item/1005005966209484.html>

<https://www.aliexpress.com/item/1005006023619761.html>

<https://www.aliexpress.com/item/1005006117321692.html>

<https://www.aliexpress.com/item/1005006040014352.html>

<https://www.aliexpress.com/item/1005006025426765.html>

Almost identical to ESPC2-12.

Documentation

[https://www.espressif.com/en/support/documents/technical-documents?field_type_tid\[\]=956](https://www.espressif.com/en/support/documents/technical-documents?field_type_tid[]=956)

Relevant documents are *ESP8684 Datasheet*, *ESP8684 Technical Reference Manual*, and *ESP32-C2 Series SoC Errata*.

ESP32-C3 and ESP8685

Official development board: ESP32-C3-DevKitM-1

Purchase link: <https://www.aliexpress.com/item/1005003989099547.html>

Documentation: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32c3/hw-reference/esp32c3/user-guide-devkitm-1.html>

Official development board: ESP32-C3-DevKitC-02

Purchase link: <https://www.aliexpress.com/item/1005004443594655.html>

Documentation: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32c3/hw-reference/esp32c3/user-guide-devkitc-02.html>

Third-party development board: WeAct Studio ESP32-C3FH4

Purchase link: <https://www.aliexpress.com/item/1005004960064227.html>

Documentation: <https://github.com/WeActStudio/WeActStudio.ESP32C3CoreBoard>

Third-party development board: Muse Lab nanoESP32-C3

Purchase link: <https://www.aliexpress.com/item/1005003081928629.html>

Documentation: <https://github.com/wuxx/nanoESP32-C3>

Third-party development board: XIAO ESP32C3

Purchase link: <https://www.aliexpress.com/item/33011482127.html>

Documentation: https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/

Third-party development board: VCC-GND YD-ESP32-C3

Purchase links:

<https://www.aliexpress.com/item/1005005242419250.html>

<https://www.aliexpress.com/item/1005006141717662.html>

Documentation: <http://www.vcc-gnd.com/>

Breaks the ESP32-C3's USB interface out and provides a separate USB-to-serial interface.

Third-party development board: ESP32C3-MINI-DK

Purchase links:

<https://www.aliexpress.com/item/1005004994621831.html>

<https://www.aliexpress.com/item/1005004945580114.html>

<https://www.aliexpress.com/item/1005004945500567.html>

Has similar features as the [ESP32-C3-DevKitC-02](#) except uses ESP32-C3-MINI-1 instead of ESP32-C3-WROOM-02 and CH340 instead of CP2102, and costs half the price.

Third-party development board: ESP32-C3 Mini

Purchase links:

<https://www.aliexpress.com/item/1005005780121305.html>

<https://www.aliexpress.com/item/1005005757810089.html>

<https://www.aliexpress.com/item/1005005692188666.html>

Ultra-miniature development board with 16 pins and a ceramic antenna.

Documentation: <http://wiki.icbbuy.com/doku.php?id=developmentboard:esp32-c3mini>

Third-party development board: 01Space ESP32-C3FH4-RGB

Purchase links:

<https://www.aliexpress.com/item/1005005872253063.html>

<https://www.aliexpress.com/item/1005005037654381.html>

<https://www.aliexpress.com/item/1005005377159331.html>

Ultra-miniature development board with ceramic antenna and 25 x WS2812 on the back.

GitHub: <https://github.com/01Space/ESP32-C3FH4-RGB>

Third-party development board: Luatos ESP32C3-CORE

Purchase links:

<https://www.aliexpress.com/item/1005004848961564.html>

<https://www.aliexpress.com/item/1005004477968500.html>

<https://www.aliexpress.com/item/1005004490215444.html>

<https://www.aliexpress.com/item/1005004539320086.html>

<https://www.aliexpress.com/item/1005004496815485.html>

Documentation: <https://wiki.luatos.com/chips/esp32c3/board.html>

Documentation

[https://www.espressif.com/en/support/documents/technical-documents?field_type_tid\[\]=785](https://www.espressif.com/en/support/documents/technical-documents?field_type_tid[]=785)

Relevant documents are *ESP32-C3 Datasheet*, *ESP32-C3 Technical Reference Manual*, and *ESP32-C3 Series SoC Errata*.

ESP32-C6

Official development board: ESP32-C6-DevKitM-1

Purchase link: <https://www.aliexpress.com/item/1005005087127863.html>

Documentation: <https://docs.espressif.com/projects/espressif-esp-dev-kits/en/latest/esp32c6/esp32-c6-devkitm-1/>

Official development board: ESP32-C6-DevKitC-1

Purchase link: <https://www.aliexpress.com/item/1005005087160183.html>

Documentation: <https://docs.espressif.com/projects/espressif-esp-dev-kits/en/latest/esp32c6/esp32-c6-devkitc-1/>

Third-party development board: WeAct Studio ESP32-C6-A

Purchase link: <https://www.aliexpress.com/item/1005005569520224.html>

Compatible with ESP32-C6-DevKitC-1.

GitHub: <https://github.com/WeActStudio/WeActStudio.ESP32-C6-A>

Third-party development board: WeAct Studio ESP32-C6-MINI

Purchase link: <https://www.aliexpress.com/item/1005006800070921.html>

GitHub: <https://github.com/WeActStudio/WeActStudio.ESP32C6-MINI>

Third-party development board: Muse Lab nanoESP32-C6

Purchase link: <https://www.aliexpress.com/item/1005005508686571.html>

GitHub: <https://github.com/wuxx/nanoESP32-C6>

Third-party development board: Wireless-Tag WT9932C6

Purchase links:

<https://www.aliexpress.com/item/1005006124846794.html>

<https://www.aliexpress.com/item/1005006125493561.html>

Documentation: <https://www.wireless-tag.com/product-item-57.html> (a PDF is available on the 资料下载 tab).

Third-party development board: 01Space ESP32-C6

Purchase link: <https://www.aliexpress.com/item/1005005411761185.html>

Third-party development board: QSZNTEC WIFI6 ESP32-C6

Purchase links:

<https://www.aliexpress.com/item/1005006065011799.html>

<https://www.aliexpress.com/item/1005006033865837.html>

<https://www.aliexpress.com/item/1005006108075658.html>

<https://www.aliexpress.com/item/1005006116350216.html>

<https://www.aliexpress.com/item/1005006112185895.html>

WARNING! The pin header's pitch is **1.27mm**, this board is **unsuitable** for breadboarding.

Documentation

Data sheets & reference manuals:

[https://www.espressif.com/en/support/documents/technical-documents?field_type_tid\[\]=1177](https://www.espressif.com/en/support/documents/technical-documents?field_type_tid[]=1177)

Relevant documents are *ESP32-C6 Datasheet*, *ESP32-C6 Technical Reference Manual*.

ESP32-H2

Official development board: ESP32-H2-DevKitM-1

Purchase link: <https://www.aliexpress.com/item/1005005252175587.html>

Documentation: <https://docs.espressif.com/projects/espressif-esp-dev-kits/en/latest/esp32h2/esp32-h2-devkitm-1/>

Third-party development board: TZT ESP32-H2-DevKitM

Purchase links:

<https://www.aliexpress.com/item/1005006252422191.html>

<https://www.aliexpress.com/item/1005006135352922.html>

<https://www.aliexpress.com/item/1005006136229394.html>

<https://www.aliexpress.com/item/1005006136160905.html>

<https://www.aliexpress.com/item/1005006138562724.html>

Third-party development board: WeAct Studio ESP32H2-N4

Purchase link: <https://www.aliexpress.com/item/1005006229313190.html>

Documentation

[https://www.espressif.com/en/support/documents/technical-documents?field_type_tid\[\]=1211](https://www.espressif.com/en/support/documents/technical-documents?field_type_tid[]=1211)

Relevant documents are *ESP32-H2 Datasheet*, and *ESP32-H2 Technical Reference Manual*.

GigaDevice

Chinese name: 兆易创新 – Pinyin: zhào yì chuàngxīn – <https://www.gd32mcu.com/en>

GD32VF103

The GD32VF103 is an interesting part based on the Nuclei N200 "[Bumblebee](#)" core. As its name implies, it is intended as a RISC-V equivalent of the famous STM32F103. As of March 2024, it is out of stock everywhere, but development boards can still be found.

Official development board: GD32VF103C-START

Purchase link: <https://www.lcsc.com/product-detail/C432220.html>

Note: this evaluation board includes GigaDevice's GDLink programmer.

Third-party development board: Sipeed Longan Nano

Purchase links:

<https://www.aliexpress.com/item/1005002542610332.html>

<https://www.aliexpress.com/item/1005003467064600.html>

Documentation: <https://github.com/sipeed/Longan-DOC>

Downloads: <https://dl.sipeed.com/shareURL/LONGAN/Nano>

Note: an additional JTAG adapter is needed to program the chip. If you don't already have one (e.g. JLink), you can buy one of Sipeed's USB-JTAG/TTL.

Purchase links:

<https://www.aliexpress.com/item/1005002837417966.html>

<https://www.aliexpress.com/item/1005005349921473.html>

<https://www.aliexpress.com/item/1005002715166088.html>

<https://www.aliexpress.com/item/1005002105074475.html>

<https://www.aliexpress.com/item/1005005955727980.html>

<https://www.aliexpress.com/item/1005005958635961.html>

<https://www.aliexpress.com/item/1005005955275482.html>

Third-party development board: LILYGO TTGO T-Display-GD32

Purchase links:

<https://www.aliexpress.com/item/4000598356310.html>

<https://www.aliexpress.com/item/4000614228180.html>

<https://www.aliexpress.com/item/4000869968817.html>

<https://www.aliexpress.com/item/1005003332019963.html>

<https://www.aliexpress.com/item/1005006063187426.html>

Documentation: <https://github.com/Xinyuan-LilyGO/LilyGO-T-DisplayGD32>

Documentation, SDK and tools

Download URL: <https://www.gd32mcu.com/en/download?kw=GD32VF1>

Data sheets & reference manuals: relevant documents are *GD32VF103xx Datasheet* and *GD32VF103 User Manual*.

SDK: relevant archive is *GD32VF103 Firmware Library*.

GD32VW553

The GD32VW553 is a wireless MCU based on [Nuclei's N307 core](#) (RV32IMACFDBP). It was announced in November 2023, so not available yet on the market as of March 2024. Technical documentation is already available:

- Data sheet: <https://gd32mcu.com/en/download/5?kw=GD32VW5>
- Reference manual: <https://gd32mcu.com/en/download/6?kw=GD32VW5>

Renesas

Web site: <https://www.renesas.com/>

Renesas is the first major semi-conductor manufacturer outside China to offer RISC-V MCU.

Renesas's first RISC-V MCU are based on [Andes Technology](#) cores (N22 and D25F), but they later developed their own core, which indicates they are very serious about their RISC-V product range and that we should expect further developments.

Product range

Part	CPU	Core	MHz	Flash	RAM	Voltage	Notes
R9A02G020	RV32IMAC	Andes N22	32	48K	16K	2.7~5.5	Ultra-low-power, motor control
R9A02G021	RV32IMACB	Renesas	48	128K	16K	1.6~5.5	Ultra-low-power
R9A06G150	RV32IMAFCP	Andes D25F	100	256K	128K	2.7~3.6	Voice command and control

Renesas's RISC-V MCU are available through usual distributors such as Mouser. However, they come in QFN packages only, which are not very hobbyist-friendly.

As of June 2024, the only available development boards are those of Renesas (no third-party boards).

R9A02G020

Data sheet: <https://www.renesas.com/us/en/document/dst/r9a02g020-datasheet>

Development board: <https://www.renesas.com/us/en/products/microcontrollers-microprocessors/risc-v/r9a02g020-evk-r9a02g020-assp-easy-motor-control-kit>

R9A02G021

Data sheet: <https://www.renesas.com/us/en/document/dst/r9a02g021-datasheet>

Reference manual: <https://www.renesas.com/us/en/document/mah/r9a02g021-users-manual-hardware>

Development board: <https://www.renesas.com/us/en/products/microcontrollers-microprocessors/risc-v/fpb-r9a02g021-fpb-r9a02g021-risc-v-mcu-fast-prototyping-board>

R9A06G150

Data sheet: <https://www.renesas.com/us/en/document/dst/r9a06g150-datasheet>

Development board: <https://www.renesas.com/us/en/products/microcontrollers-microprocessors/risc-v/tw001-vui-riscvpocz-r9a06g150-assp-easy-voice-hmi-kit>

SOPHGO

Chinese name: 算能京 – Pinyin: suàn néng jīng – <https://en.sophgo.com/>

Also known as CVITEK (晶视智能 – jīng shì zhì néng) and SOPHON (算丰 – suàn fēng).

CV1800B

The CV1800B includes two C906 cores, one fully-fledged (RV64GCV, with V being v0.7.1), the other stripped down (no V extension, 700MHz instead of 1GHz). It also integrates 64MB RAM.

Third-party development board: Milk-V Duo

Being Linux-capable, the Milk-V Duo is comparable to some extent to the Ox64, including price-wise, though the BL808 of the Ox64 might be better suited for low-power applications.

The Milk-V Duo also comes with FreeRTOS support (included in duo-buildroot-sdk).

Note: as of 2023-08-03, a significant documentation effort has been made, including English versions, which bodes well for the future, including for other Milk-V products.

Purchase links:

<https://www.aliexpress.com/item/1005005699023966.html>

<https://www.aliexpress.com/item/1005005699176591.html>

<https://www.aliexpress.com/item/1005005699215618.html>

<https://www.aliexpress.com/item/1005005866947535.html>

<https://www.aliexpress.com/item/1005005963211206.html>

Documentation & SDK:

<https://milkv.io/docs/duo> (start here)

<https://milkv.io/duo> (see pinout at bottom of page)

<https://github.com/milkv-duo> (duo-files, hardware, duo-buildroot-sdk)

<https://github.com/milk-v> (cvitek-host-tools)

<https://community.milkv.io/c/duo/5> (forum)

Release date: Apr. 2023

SG2002

The SG2002 incorporates the same C906 and 8051 cores as the CV1800B, plus an **ARM Cortex-A53 core** running at 1GHz, and 256MB RAM.

Third-party development board: Milk-V Duo 256M

Purchase links:

<https://www.aliexpress.com/item/1005006466621314.html>

<https://www.aliexpress.com/item/1005006444164705.html>

<https://www.aliexpress.com/item/1005006442665261.html>

<https://www.aliexpress.com/item/1005006444438676.html>

Documentation and SDK share the same repositories as the Milk-V Duo.

Release date: Jan. 2024

Third-party development board: Sipeed LicheeRV Nano

Purchase link: <https://www.aliexpress.com/item/1005006519668532.html>

Documentation: http://dl.sipeed.com/shareURL/LICHEE/LicheeRV_Nano

Release date: Jan. 2024

SG2000

The SG2000 is similar to the SG2002 but offers 512MB RAM.

Third-party development board: Milk-V Duo S

Purchase links:

<https://www.aliexpress.com/item/1005006719645936.html>

<https://www.aliexpress.com/item/1005006975222408.html>

<https://www.aliexpress.com/item/1005007013659841.html>

<https://www.aliexpress.com/item/1005007377192297.html>

Documentation and SDK share the same repositories as the Milk-V Duo.

WCH

Chinese name: 南京沁恒 (or just 沁恒) – Pinyin: nánjīng qìn héng – <https://wch-ic.com/>

WCH provides a much wider range of RISC-V MCU than any other manufacturer. For this reason, each chip has its own section below.

WCH's offer is divided in 2 families, the general purpose CH32Vxxx series, which are the RISC-V equivalents of their ARM CH32Fxxx series, and the more specialised CHxxx series, which use a slightly different peripheral set.

An interesting thing to note is the CH32V003 doesn't have an ARM equivalent, which could be a hint that, like Espressif, WCH will continue to concentrate their efforts on their RISC-V products.

The following table presents the main differences between WCH's RISC-V MCU:

Part+GitHub	CPU	Core	MHz	Flash	RAM	Voltage	Notes
CH32V002	RV32mEC	V2C	48	16K	4K	1.9-5.5V	Low pin count (8, 16, 20)
CH32V003	RV32EC	V2A	48	16K	2K	2.7-5.5V	Low pin count (8, 16, 20)
CH32V004	RV32mEC	V2C	48	32K	6K	1.9-5.5V	Low pin count (20)
CH32V005	RV32mEC	V2C	48	32K	6K	1.9-5.5V	Low pin count (20)
CH32V006	RV32mEC	V2C	48	62K	8K	1.9-5.5V	Low pin count (20)
CH32V007	RV32mEC	V2C	48	62K	8K	1.9-5.5V	
CH32M007	RV32mEC	V2C	48	62K	8K	1.8-5.0V	Motor control, includes MOSFET drivers
CH32M030	RV32IMCB	V3B	72	64K	12K	5-2.8V	Motor control, includes MOSFET drivers
CH32X035	RV32IMAC	V4C	48	62K	20K	2.0-5.5V	USB PD, PIOC, OpAmp/PGA/comp.
CH32V103	RV32IMAC	V3A	80	64K	20K	2.7-5.5V	
CH32L103	RV32IMAC	V4C	96	64K	20K	2.4-3.6V	Low-power, USBPD, OpAmp/PGA/comp.
CH32V203	RV32IMAC	V4B	144	128K	64K	2.4-3.6V	Low-power. LQFP-64: 10M Ethernet
CH32V208	RV32IMAC	V4C	144	128K	64K	2.4-3.6V	BLE, 10M Ethernet
CH32V303	RV32IMAFc	V4F	144	256K	64K	2.4-3.6V	
CH32V305	RV32IMAFc	V4F	144	128K	32K	2.4-3.6V	
CH32V307	RV32IMAFc	V4F	144	256K	64K	2.4-3.6V	10M Ethernet. LQFP-100: DVP&FSMC
CH32V317	RV32IMAFc	V4F	144	256K	64K	2.4-3.6V	= CH32V307 but with 10M/100M Eth.
CH564	RV32IMAC	V4J	120	192K	128K	3.3 / 5V	USB HS, USB PD, 10M/100M Ethernet
CH565	RV32IMAC	V3A	120	448K	96K	2.3-3.6V	USB 3.0, 1000M Ethernet, EMMC, DVP
CH569	RV32IMAC	V3A	120	448K	96K	2.3-3.6V	USB 3.0, 1000M Ethernet, EMMC, HSPI
CH573/571	RV32IMAC	V3A	60	448K	16K	2.3-3.6V	Low-power, BLE
CH583/582	RV32IMAC	V4A	80	448K	30K	2.3-3.6V	Low-power, long-range BLE
CH585/584	RV32IMCB	V3C	78	448K	96K	1.7-3.6V	BLE, NFC, LCD, LED, USB HS
CH592/591	RV32IMAC	V4C	80	448K	24K	2.3-3.6V	Low-power, BLE, LCD
CH641	RV32EC	V2A	48	16K	2K	4-12.6V	USB PD, USB BC
CH643	RV32IMAC	V4C	48	62K	20K	2.0-5.5V	USB PD, PIOC, RGB LED PWM
CH645	RV32IMAC	V4C	125	224K	80K		USB HUB, KVM, 10M/100M Ethernet

PIOC: programmable I/O protocol controller / PGA: programmable gain amplifier

RV32EmC = RV32EC_Zmmul

Note: Development boards for the CH573 have intentionally been omitted as CH582 is preferred.

For further details, see selection table covering all WCH MCU: <https://special.wch.cn/en/mcu/>, or the page dedicated to their RISC-V MCU: https://special.wch.cn/zh_cn/RISCV_MCU_Index/

Note: all WCH MCU SDK include the schematics of the official evaluation boards as reference designs.

GitHub: <https://github.com/openwch>

YouTube: <https://www.youtube.com/@rvmaking/videos> (tutorials, tips and tricks...)

Programmer / debugger

A proprietary programmer / debugger is needed to flash WCH chips, the WCH-LinkE. It includes both the programmer/debugger, and a USB-to-serial adapter, eliminating the need for a separate device.

Purchase links:

<https://www.aliexpress.com/item/1005004881582037.html>

<https://www.aliexpress.com/item/1005004964197577.html>

<https://www.aliexpress.com/item/1005005983875152.html>

<https://www.aliexpress.com/item/1005005244468643.html>

One is already included with the CH32V003 evaluation kit, and the CH32V003 + CH32V203 evaluation kit combo from WCH.

Note: you may also come across the WCH-Link (without final E) on AliExpress. It is the predecessor of the WCH-LinkE and does not support the CH32V003/CH32X035/CH643.

Documentation:

User manual: https://www.wch-ic.com/downloads/WCH-LinkUserManual_PDF.html

Schematic diagram: https://www.wch.cn/downloads/WCH-LinkSCH_PDF.html

JTAG upgrade utility for WCH-LinkE-R0-1v3:

https://www.wch.cn/downloads/WCHLinkEJtagUpdTool_ZIP.html

Working mode

The WCH-LinkE and WCH-Link have 2 distinct working modes, ARM and RISC-V. When the blue LED is on, the device is in ARM mode and cannot be used with RISC-V MCU.

The WCH-LinkE has a "ModeS" push button to toggle the working mode. Remove the transparent plastic case, and hold "ModeS" down while plugging the device in a USB port, this will change the working mode and save it so you don't have to repeat the operation the next time you use it.

Older WCH-Link don't have this button. To toggle the working mode, you need to short TX to GND while plugging the device in a USB port. The new mode will also be saved.

Flashing utility

WCH provides 3 flashing tools, WCHISPTool (Windows application, recommended), WCHISPTool_CMD (multi-platform command-line tool), and WCH-LinkUtility (Windows application). MounRiver Studio includes its own flashing utility, which looks a lot like WCH-LinkUtility.

Download links:

https://www.wch.cn/downloads/WCHISPTool_Setup_exe.html

https://www.wch.cn/downloads/WCHISPTool_CMD_ZIP.html

https://www.wch.cn/downloads/WCH-LinkUtility_ZIP.html

MounRiver IDE

MounRiver Community Edition is an Eclipse-based IDE supporting all RISC-V and ARM WCH MCU, available for Windows, Linux and Mac.

Download link: <http://www.mounriver.com/download>

Stand alone tool chain

WCH provides modified versions of OpenOCD and GCC to support their MCU's specific features. These are included with MounRiver, but can also be downloaded separately through MounRiver's download page, e.g. for installation on a continuous integration server.

Alternatives to WCH tools

If you would like to use mainstream GCC instead of WCH's, or to try an open-source flashing tool, the following Reddit threads will be of interest to you:

<https://www.reddit.com/r/RISCV/comments/115u6i9/comment/j94xvpq/>

<https://www.reddit.com/r/RISCV/comments/126262j/>

Note to Linux users

You can find detailed explanations on how to install and use WCH's development tools on Linux in this document:

https://codeberg.org/20-100/Awesome_RISC-V/src/branch/master/WCH/WCH_on_Linux.pdf

CH32L103

Official development board

Purchase link: <https://www.aliexpress.com/item/1005006671545123.html>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32L103DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32L103RM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32L103EVT_ZIP.html

CH32V003

Official development board

Purchase link: <https://www.aliexpress.com/item/1005004895791296.html>

Note: this board is equipped with a crystal, so GPIO pins PA1 and PA2 are not available.

Third-party development board: CR-CH32VXX

Purchase links:

<https://www.aliexpress.com/item/1005005878452720.html>
<https://www.aliexpress.com/item/1005005879547898.html>
<https://www.aliexpress.com/item/1005005870406150.html>
<https://www.aliexpress.com/item/1005005871348984.html>
<https://www.aliexpress.com/item/1005005910184015.html>

Note: this board doesn't have a crystal, so GPIO pins PA1 and PA2 are available.

Third-party development board: WeAct Studio CH32V003F4U6

This board exists in 2 variants, 3.3V and 5V, to be selected according to the supply voltage of the peripherals you wish to connect.

Purchase link: <https://www.aliexpress.com/item/1005006217778264.html>

Documentation: <https://github.com/WeActStudio/WeActStudio.CH32V003CoreBoard>

Note: this board doesn't have a crystal, so GPIO pins PA1 and PA2 are available.

Third-party development board: Liludin CH32V003 V1772

Purchase links:

<https://www.aliexpress.com/item/1005008012887438.html>
<https://www.aliexpress.com/item/1005008029047627.html>
<https://www.aliexpress.com/item/1005008105449396.html>
<https://www.aliexpress.com/item/1005006781371488.html>
<https://www.aliexpress.com/item/1005008103356416.html>

Note: this board doesn't have a crystal, so GPIO pins PA1 and PA2 are available.

Third-party development board: BTE23-15

This board exists in 2 variants, 3.3V and 5V, to be selected according to the supply voltage of the peripherals you wish to connect.

Purchase link: <https://www.aliexpress.com/item/1005005901472089.html>

Note: this board is equipped with a crystal, so GPIO pins PA1 and PA2 are not available.

Third-party development board: Muse Lab nanoCH32V003

Purchase links:

<https://www.aliexpress.com/item/1005005221751705.html>
<https://www.aliexpress.com/item/1005005222228477.html>

Documentation: <https://github.com/wuxx/nanoCH32V003>

Note: this board is equipped with a crystal, so GPIO pins PA1 and PA2 are not available.

Third-party development board: QSZNTEC CH32V003

Purchase links:

<https://www.aliexpress.com/item/1005004964355080.html>

<https://www.aliexpress.com/item/1005005137124754.html>

Note: this board is equipped with a crystal, so GPIO pins PA1 and PA2 are not available.

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V003DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32V003RM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV2_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V003EVT_ZIP.html

CH32V103

Official development board

Purchase links:

<https://www.lcsc.com/product-detail/C2943983.html> (C8T6)

<https://www.lcsc.com/product-detail/C2943982.html> (R8T6)

<https://www.aliexpress.com/item/1005004607642695.html> (C8T6, R8T6)

Third-party development board: Shenzhen LC Technology 303CH32VC02 = CH32V103C_MINI

Purchase links:

<https://www.aliexpress.com/item/1005005772209605.html>

<https://www.aliexpress.com/item/1005007948183240.html>

<https://www.aliexpress.com/item/1005005246058814.html>

<https://www.aliexpress.com/item/1005005226811776.html>

<https://www.aliexpress.com/item/1005005138899141.html>

Third-party development board: Shenzhen LC Technology 303CH32MI01 = CH32V103R_MINI

Purchase links:

<https://www.aliexpress.com/item/1005005768751075.html>

<https://www.aliexpress.com/item/1005006079067376.html>

<https://www.aliexpress.com/item/1005005218858522.html>

<https://www.aliexpress.com/item/1005004569522706.html>

<https://www.aliexpress.com/item/1005005786828189.html>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V103DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32xRM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV3_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V103EVT_ZIP.html

CH32V203

Official development boards

Purchase links:

<https://www.aliexpress.com/item/1005004493040662.html> (CH32V203C8T6)

<https://www.aliexpress.com/item/1005005335685988.html> (F6P6, F8P6, G6U6, G8R6)

Third-party development board: Muse Lab nanoCH32V203

Purchase links:

<https://www.aliexpress.com/item/1005004908206775.html>

<https://www.aliexpress.com/item/1005006127448361.html>

<https://www.aliexpress.com/item/1005006125935583.html>

<https://www.aliexpress.com/item/1005006128106996.html>

<https://www.aliexpress.com/item/1005005065054068.html>

Documentation: <https://github.com/wuxx/nanoCH32V203>

Third-party development board: WeAct Studio BluePill+ CH32V203

Purchase link: <https://www.aliexpress.com/item/1005001474741936.html>

Documentation: <https://github.com/WeActStudio/WeActStudio.BluePill-Plus-CH32>

Note: this board initially shipped with a CH32V103C8T6, but now uses a CH32V203C8T6.

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V203DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32FV2x_V3xRM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V20XEVT_ZIP.html

CH32V208

Official development board

Purchase link: <https://www.aliexpress.com/item/1005004924242063.html>

Third-party development board: Shenzhen LC Technology 303CH32ETH0 = CH32V208_ETH

Purchase links:

<https://www.aliexpress.com/item/1005007257716209.html>

<https://www.aliexpress.com/item/1005007287667154.html>

<https://www.aliexpress.com/item/1005007305434082.html>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V208DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32FV2x_V3xRM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V20XEVT_ZIP.html

CH32V303

Official development board

Purchase link: <https://www.aliexpress.com/item/1005005444077007.html>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V307DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32FV2x_V3xRM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V307EVT_ZIP.html

CH32V305

Official development board

Purchase links:

<https://www.aliexpress.com/item/1005005444077007.html> (FBP6 & RBT6)

<https://www.aliexpress.com/item/1005006409708734.html> (FBP6)

Third-party development board: Muse Lab nanoCH32V305

Purchase links:

<https://www.aliexpress.com/item/1005005033298927.html>

<https://www.aliexpress.com/item/1005005705171817.html>

<https://www.aliexpress.com/item/1005005180667965.html>

Documentation: <https://github.com/wuxx/nanoCH32V305>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V307DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32FV2x_V3xRM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V307EVT_ZIP.html

CH32V307

Official development board

Purchase links:

<https://www.aliexpress.com/item/1005005444077007.html> (RCT6 & WCU6)

<https://www.aliexpress.com/item/1005004329125620.html> (VCT6)

<https://www.lcsc.com/product-detail/C2943980.html> (VCT6)

Third-party development board: VCC-GND YD-CH32V307VC

Purchase links:

<https://www.aliexpress.com/item/1005005882943775.html>

<https://www.aliexpress.com/item/1005005871739964.html>

<https://www.aliexpress.com/item/1005005887620227.html>

<https://www.aliexpress.com/item/1005005933803835.html>

Documentation: <http://www.vcc-gnd.com/>

Third-party development board: VCC-GND YD-CH32V307RC

Purchase link: <https://www.aliexpress.com/item/1005005175711704.html>

Third-party development board: VCC-GND CH32V307RC-MINI

Purchase link: <https://www.aliexpress.com/item/1005005175678285.html>

Third-party development board: VCC-GND CH32V307VCT6-MINI

Purchase link: <https://www.aliexpress.com/item/1005005175835038.html>

Third-party development board: CH32V307RCT6-1.0

Purchase link: <https://www.aliexpress.com/item/1005005616868638.html>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32V307DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32FV2x_V3xRM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32V307EVT_ZIP.html

CH32X035

Official development board

Purchase link: <https://www.aliexpress.com/item/1005005718558442.html>

Third-party development board: WeAct Studio CH32X035F8U6

Purchase link: <https://www.aliexpress.com/item/1005006909948695.html>

Documentation: <https://github.com/WeActStudio/WeActStudio.CH32X035CoreBoard>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH32X035DS0_PDF.html

Reference manual: http://wch-ic.com/downloads/CH32X035RM_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH32X035EVT_ZIP.html

CH565

Official development board

Purchase link: <https://www.aliexpress.com/item/1005004346104186.html>

Documentation and SDK

Same as CH569.

CH569

Official development board

Purchase links:

<https://www.lcsc.com/product-detail/C3001176.html>
<https://www.aliexpress.com/item/1005004328816871.html>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH569DS1_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV3_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH569EVT_ZIP.html

CH582 / CH583

Official development board

Purchase links:

<https://www.aliexpress.com/item/1005005060737000.html> (CH582M and CH583M)
<https://www.aliexpress.com/item/1005005493310632.html> (CH583M)
<https://www.aliexpress.com/item/1005004346585597.html> (CH582F)
<https://www.lcsc.com/product-detail/C2943981.html> (CH582M)

Third-party development board: VCC-GND YD-CH58x (CH582M)

Purchase links:

<https://www.aliexpress.com/item/1005005305938011.html>
<https://www.aliexpress.com/item/1005005609370215.html>
<https://www.aliexpress.com/item/1005005305272257.html>
<https://www.aliexpress.com/item/1005005305557552.html>
<https://www.aliexpress.com/item/1005005917109661.html>

Documentation: <http://www.vcc-gnd.com/>

Third-party development board: Shenzhen LC Technology 303CH582M01 = CH582M

Purchase links:

<https://www.aliexpress.com/item/1005005245945123.html>
<https://www.aliexpress.com/item/1005005457754241.html>
<https://www.aliexpress.com/item/1005005456987838.html>

Similar to YD-CH58x. After ordering, ask the seller for schematic diagram. Uses the USB C connector for programming, which at the moment only works under Windows.

Third-party development board: WeAct Studio CH582F

Purchase links:

<https://www.aliexpress.com/item/1005004784988010.html>

<https://www.aliexpress.com/item/1005004870340829.html>

<https://www.aliexpress.com/item/1005004909931218.html>

<https://www.aliexpress.com/item/1005005794826634.html>

Documentation: <https://github.com/WeActStudio/WeActStudio.WCH-BLE-Core>

Documentation and SDK

Data sheet: http://wch-ic.com/downloads/CH583DS1_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH583EVT_ZIP.html

CH584 / CH585

Official development board

Purchase links:

<https://www.aliexpress.com/item/1005007883604872.html>

<https://www.aliexpress.com/item/1005008035217856.html>

<https://www.aliexpress.com/item/1005007884229301.html>

CH592 / CH591

Official development board

Purchase link: <https://www.aliexpress.com/item/1005005884261132.html>

Third-party development board: WeAct Studio CH592F

Purchase link: <https://www.aliexpress.com/item/1005006117859297.html>

Documentation: <https://github.com/WeActStudio/WeActStudio.WCH-BLE-Core>

Documentation and SDK

Data sheet: https://wch-ic.com/downloads/CH592DS1_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV4_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH592EVT_ZIP.html

CH641

Official development board

Purchase link: <https://www.aliexpress.com/item/1005006289346429.html>

Documentation and SDK

Data sheet: https://wch-ic.com/downloads/CH641DS0_PDF.html

Processor manual: http://wch-ic.com/downloads/QingKeV2_Processor_Manual_PDF.html

SDK: https://www.wch.cn/downloads/CH641EVT_ZIP.html

Before you read on, a few words of warning

The following chapters provides information about chips whose manufacturer apparently consider domestic-only.

This means that you won't be able to download development tools and official documentation, which are offered in developer portals where registration often requires a Chinese mobile phone number, or use captchas with Chinese characters.

Anyway, said documentation is entirely in Chinese, so if you're not fluent in Chinese, you may feel discouraged by the task of navigating through such a volume of text and translating the parts you need. Google Translate works well, but it's a big task.

You won't be able to purchase development boards directly from the manufacturer either. Many have a Taobao store, but you may not have a Taobao account.

It is usually possible to find blog posts on Chinese web sites providing guidance for your first steps. Translating them is manageable, but their informative content is limited compared to the often unavailable official documentation.

You can purchase some of these chips from LCSC and/or find development boards on AliExpress, but again, you would be luckier on Taobao.

Before you buy a development board, it is recommended to ask the seller if they can provide you with documentation. Most of the time, supporting material is made available on Baidu, which cannot be used outside of China. Some sellers provide these materials on download sites accessible from the rest of the world, but not all do.

Some third-party board manufacturers have a web site, so you can also ask them directly, or possibly even order from there.

Bluetrum

Chinese name: 中科蓝讯 – Pinyin: zhōng kē lán xùn – <https://www.bluetrum.com/>

AB5301A

Official development board: AB32VG1 "blue board" (aka. prougen)

Purchase links:

<https://www.aliexpress.com/item/1005003476403583.html>

<https://www.aliexpress.com/item/1005003569918832.html>

<https://www.aliexpress.com/item/1005003124829942.html>

Third-party development board: AB32VG1 "green board"

Purchase links:

<https://www.aliexpress.com/item/1005003547381454.html>

<https://www.aliexpress.com/item/1005003700027776.html>

<https://www.aliexpress.com/item/1005003547529433.html>

<https://www.aliexpress.com/item/1005003619461823.html>

<https://www.aliexpress.com/item/1005003594401351.html>

Common description

The only difference between the two boards is the PCB layout, all the rest is identical. The green version has the advantage of being much cheaper and more easily available than the blue version.

These boards are available through AliExpress, but their documentation is only partially available in English, and they are supported by the [RT-Thread](#) RTOS.

This means that these boards are not for beginners, and you must be willing to [Google Translate](#) some Chinese documents and/or blog posts, but provided you have a first experience with another Bluetooth MCU, you should not hit major roadblocks.

Documentation & SDK

<https://github.com/BLUETRUM>

https://gitee.com/bluetrum/bluetrum_sdk

<https://www.cnx-software.com/2021/09/12/getting-started-with-bluetrum-ab32vg1-risc-v-bluetooth-audio-board-using-rt-thread/>

<https://ab32vg1-example.readthedocs.io/zh/latest/>

<https://docs.qq.com/doc/DTVVWWXpLRVI6cER2>

Canaan

Chinese name: 嘉楠科技 – Pinyin: jiā nán kējì – <https://www.canaan.io/>

Canaan produces the Kendryte series, 64-bit RISC-V MCU intended for edge AI.

Canaan has the detestable habit of been quite shy about their products, so you won't be able to find a proper data sheet and technical reference manual, which all other silicon vendors proudly offer on their web sites.

They have a developer forum, but it hasn't seen a new message in the last 12 months:

<https://developer.canaan.io/>

Their GitHub account offers downloads (IDE, SDK), but no decent documentation:

<https://github.com/kendryte>

A cursory look at GitHub reveals that many people have played with these chips, so you may possibly get the information you need by browsing their repositories.

K210

The K210 features 2 RV64IMAFDC cores running at 400MHz, 8MB SRAM, a neural network processor, and an audio processor. It was released in 2018, so it's a bit old now (2023), but development boards are still available.

Superficial data sheet: <https://github.com/kendryte/kendryte-doc-datasheet>

Third-party development boards: Sipeed Maix series

Purchase links:

<https://www.aliexpress.com/item/1005002802675701.html> (Maix Amigo)

<https://www.aliexpress.com/item/1005002624234145.html> (Maix Cube)

<https://www.aliexpress.com/item/1005002569741906.html> (Maix Dock)

<https://www.aliexpress.com/item/1005002547345797.html> (Maix Duino)

<https://www.aliexpress.com/item/1005004131749651.html> (Maix Nano)

GitHub: <https://github.com/sipeed/>

Documentation: <https://mega.nz/folder/A8g1Hb4J#WcuoqvbpasKlVB8-YEpWPA/folder/4wpEgIQZ>

Third-party development boards: AI-Motion series

Purchase links:

<https://www.aliexpress.com/item/1005005475178374.html> (K210 developer kit)

<https://www.aliexpress.com/item/1005005456612901.html> (K210 visual recognition)

Manufacturer web site: <https://www.yahboom.com/>

Third-party development board: pyAI-K210

Purchase link: <https://www.aliexpress.com/item/1005001459205624.html>

Third-party development board: M5Stack M5StickV

Purchase link: <https://www.aliexpress.com/item/1005003299167263.html>

Manufacturer web site: <https://m5stack.com/>

GitHub: <https://github.com/m5stack/M5-StickV-UnitV>

K230

The K230 integrates two T-Head C908 cores, one running at 800MHz and supporting RV64GCB, and the other running at 1.6GHz with RISC-V Vector 1.0 support.

Official development board: Canaan CanMV-K230

Purchase links:

<https://www.aliexpress.com/item/1005006164536639.html>

<https://www.aliexpress.com/item/1005006164476445.html>

<https://www.analoglamb.com/product/kendryte-k230-risc-v-development-board-canmv-k230/>

GitHub: <https://github.com/kendryte> (several repositories)

K510

The K510 is introduced by Canaan as the bigger brother of the K210, but as of the 2023-08-25, I haven't been able to find even a product brief about it...

Official development board: Canaan K510 CRB-KIT

Purchase links:

<https://www.aliexpress.com/item/1005005278496505.html>

<https://www.aliexpress.com/item/1005004332478616.html>

<https://www.aliexpress.com/item/1005004255412488.html>

<https://www.aliexpress.com/item/1005004254339055.html>

GitHub: <https://github.com/kendryte> (several repositories)

Third-party development board: DongshanPI-Vision

Purchase link: <https://www.aliexpress.com/item/1005005648555879.html>

GitHub: <https://github.com/DongshanPI> (several repositories)

HiSilicon

Chinese name: 海思 – Pinyin: hǎi sī – <https://www.hisilicon.com/en>

Hi3861

HiSilicon is a subsidiary of Huawei. Their chips are supported by [HarmonyOS](#) (鸿蒙, pinyin: hóngméng), an OS developed by Huawei using the LiteOS kernel.

The Hi3861 is comparable to the ESP32-C3 or the BL602. It exists in 2 models, the Hi3861V100 and the Hi3861LV100 (low power version). It is based on an RV32IMC core running at up to 160MHz, with 352KB RAM and 2MB flash. It supports IEEE 802.11b/g/n up to 72.2 Mb/s.

Unofficial documentation (English): https://github.com/koendv/hi3861_notes

Third-party development board: Ai-Thinker Hi-12F-Kit & Hi-12FL-Kit

The 12F is based on the Hi3861V100, and the 12FL on the Hi3861LV100.

Purchase links:

<https://www.aliexpress.com/item/1005005806328088.html>

<https://www.aliexpress.com/item/1005004072457367.html>

<https://www.aliexpress.com/item/1005004116773752.html>

<https://www.aliexpress.com/item/1005005003338511.html>

<https://www.aliexpress.com/item/1005005807889439.html>

Documentation: <https://docs.ai-thinker.com/en/hi>

Third-party development board: Shenzhen LC Technology 303HI386101 = Hi3861

This board is based on the Hi3861V100 and costs twice the price of the Hi-12F(L)-Kit.

Purchase links:

<https://www.aliexpress.com/item/1005004416979325.html>

<https://www.aliexpress.com/item/1005003624801955.html>

Nanjing Zhongke Micro

Chinese name: 南京中科微 – Pinyin: nánjīng zhōng kē wēi – <https://www.njzkwiot.com/>

Previously known as CSM (<http://csm-ic.com/>).

Their products can be purchased from Taobao, including development boards.

CSM32RV003

This low-power RV32IMAC MCU can run at 32MHz, support a wide supply voltage range (1.8~5.5V), include 32KB Flash / 4KB SRAM, a fast high-precision 16-bit ADC, and provide 2-wire cJTAG support (the 'c' before 'JTAG' is not a typo).

The name may suggest this MCU intends to compete with WCH's famous CH32V003.

CSM32RV20

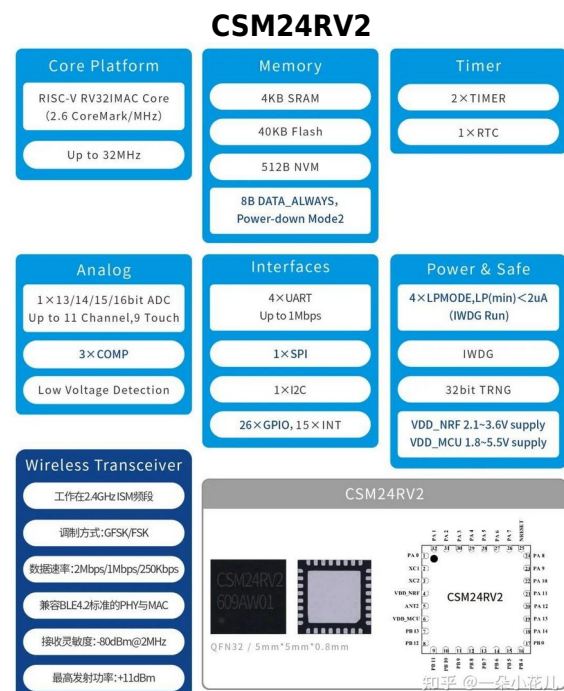
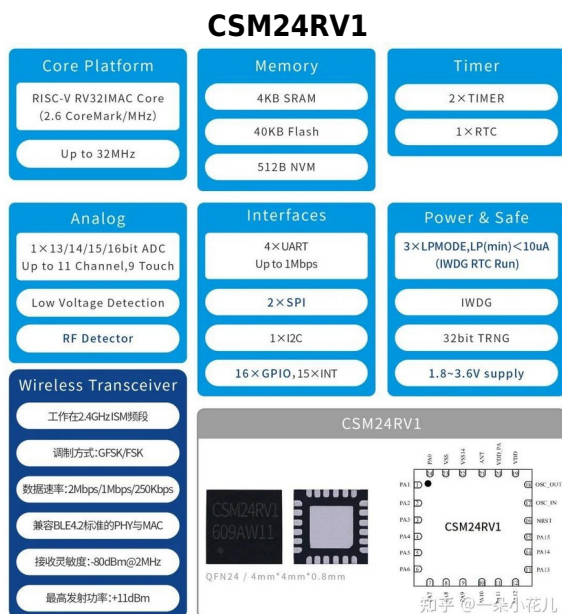
This MCU is very similar to the CSM32RV003 except that it has 40KB flash and added features. It can be purchased from LCSC, in addition to Taobao.

Data sheet & firmware library: https://github.com/cjacker/csm32rv20_firmware_library

Tutorial: <https://github.com/cjacker/opensource-toolchain-csm32rv20>

CSM24RVx series

The CSM24RV1 and CSM24RV2 integrate a RISC-V MCU and a 2.4GHz wireless transceiver. They inherit their peripherals from the CSM32RV20. These chips suggest that RISC-V is a strategic choice for CSM and is intended to replace their proprietary 8-bit RISC cores in future products.



Xinsheng Technology

Chinese name: 芯昇科技 – Pinyin: xīn shēng kējì – <https://www.xinshengcmiot.cn/>

Xinsheng Technology is a division of **China Mobile** (中国移动 – zhōngguó yídòng).

CM32M4xxR

There are 2 chips in this series, the CM32M431R and CM32M433R. The CM32M433R can be purchased from LCSC and development boards from AliExpress.

These chips are designed to compete with ARM Cortex-M3/M4/M4F and M33. They are built on an RV32IMAFDCP Nuclei N308 core. They can run at 144MHz, include up to 512KB Flash / 144KB SRAM, fast 12-bit ADC and DAC (5Mbps and 1Mbps respectively), and provide 2-wire/4-wire JTAG support.

Documentation, SDK and tools

Data sheet: <https://www.rvmcu.com/app/quickstart/skins/default/doc/CM32M4xxR-datasheet-V1.4.pdf>

Reference manual: <https://www.rvmcu.com/app/quickstart/skins/default/doc/CM32M4xxR-user-guide-V1.4.pdf>

Other documents (e.g. application notes, libraries):
<https://www.xinshengcmiot.cn/service/download>

Development tools: <https://nucleisys.com/download.php> (include toolchain, OpenOCD, and Nuclei Studio, an Eclipse-based IDE)

Official development board: CM32R433R-START

Purchase link: <https://www.aliexpress.com/item/1005004333840765.html>

Board schematic: <https://www.rvmcu.com/app/quickstart/skins/default/doc/cm32m433r-start-sch.pdf>

User manual (Chinese): <https://www.rvmcu.com/app/quickstart/skins/default/doc/CM32M433R-START-User-Manual-V1.pdf>

Unofficial documentation (English):
https://codeberg.org/20-100/Awesome_RISC-V/src/branch/master/ChinaMobile/CM32M433R.pdf